

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project: Campus Marketplace Portal (Buy & Sell Used Items)

1. Problem Understanding

In many colleges, students and staff have unused items such as books, calculators, lab equipment, electronics, hostel essentials, and bicycles. Currently, buying and selling these items happens through WhatsApp groups, notice boards, or word of mouth, making the process inefficient and unreliable.

A centralized campus marketplace is needed where authenticated college users can securely buy and sell items within the institution.

2. Target Users

Primary Users

- Students
- Teaching Staff
- Non-Teaching Staff

Secondary Users

- College Administration (Admin)
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3. Problems Faced by Target Users

Problem	Who Faces It?	Impact
No centralized platform	Students & Staff	Difficult to find buyers/sellers

Selling through WhatsApp groups	Students	Messages get buried quickly
Fake or unknown buyers	Students & Staff	Trust and security issues
No easy search	Buyers	Takes time to find products
No verification of users	Everyone	Risk of scams
No communication system	Buyers & Sellers	Must use personal phone numbers
Difficult to know if an item is sold	Buyers	Wastes time contacting sellers
No record of transactions	Users	No purchase/selling history
Manual coordination	Everyone	Time-consuming
No reporting mechanism	Users	Difficult to report fake listings

4. Proposed Solution

Develop a secure web portal exclusively for college students and staff where every authenticated user can both buy and sell used items.

The system provides:

- Secure login using institutional email
 - Product listing with images
 - Product search and filtering
 - Buyer–seller chat
 - Wishlist
 - Product status (Available/Sold)
 - Notifications
 - Admin moderation
 - Reports and analytics
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5. Objectives

- Create a trusted campus marketplace
- Reduce waste through item reuse
- Simplify buying and selling
- Improve communication between buyers and sellers
- Ensure secure and transparent transactions
- Digitize the existing manual process

6. Core Features

Module	Requirements	
User Authentication	College email login, registration, password reset	
User Profile	Name, department, year, profile photo, contact details	
Product Management	Add, edit, delete, view products with images, price, category, condition	
Product Categories	Books, Electronics, Furniture, Cycles, Hostel Essentials, Others	
Search & Filter	Search by product name, category, price, condition	
Marketplace	Browse all products and view product details	
Chat System	Buyer and seller can communicate directly	
Wishlist	Save favorite products	
Notifications	Product approval, new messages, product sold	
Admin Dashboard	Manage users, approve/reject listings, remove inappropriate products	
Reports	Report fake listings or users	
Reviews & Ratings	Buyers can rate sellers after a transaction	
Analytics	Total users, total products, sold items, active listings	
Security	Secure login, role-based access, password encryption, input validation	
Responsive Design	Works on desktop and mobile devices	

Authentication

- College email login
- User profile
- Role-based access

Marketplace

- Add product
- Edit/Delete listing
- Upload multiple images
- Product categories

- Product details page
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Search & Discovery

- Search products
 - Filter by category
 - Filter by price
 - Filter by condition
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Buyer Features

- Browse products
 - Wishlist
 - Chat with seller
 - View purchase history
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Seller Features

- Manage listings
 - Mark item as sold
 - View selling history
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Admin Features

- Manage users
 - Approve/Remove listings
 - Handle reports
 - Dashboard with analytics
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Notifications

- New messages
- Product approval
- Product sold
- Admin announcements

7. Functional Requirements

- User Registration/Login
- User Profile Management
- Product Management
- Search & Filter
- Chat System
- Wishlist
- Notifications
- Admin Dashboard
- Reporting System
- Analytics Dashboard

8. Non-Functional Requirements

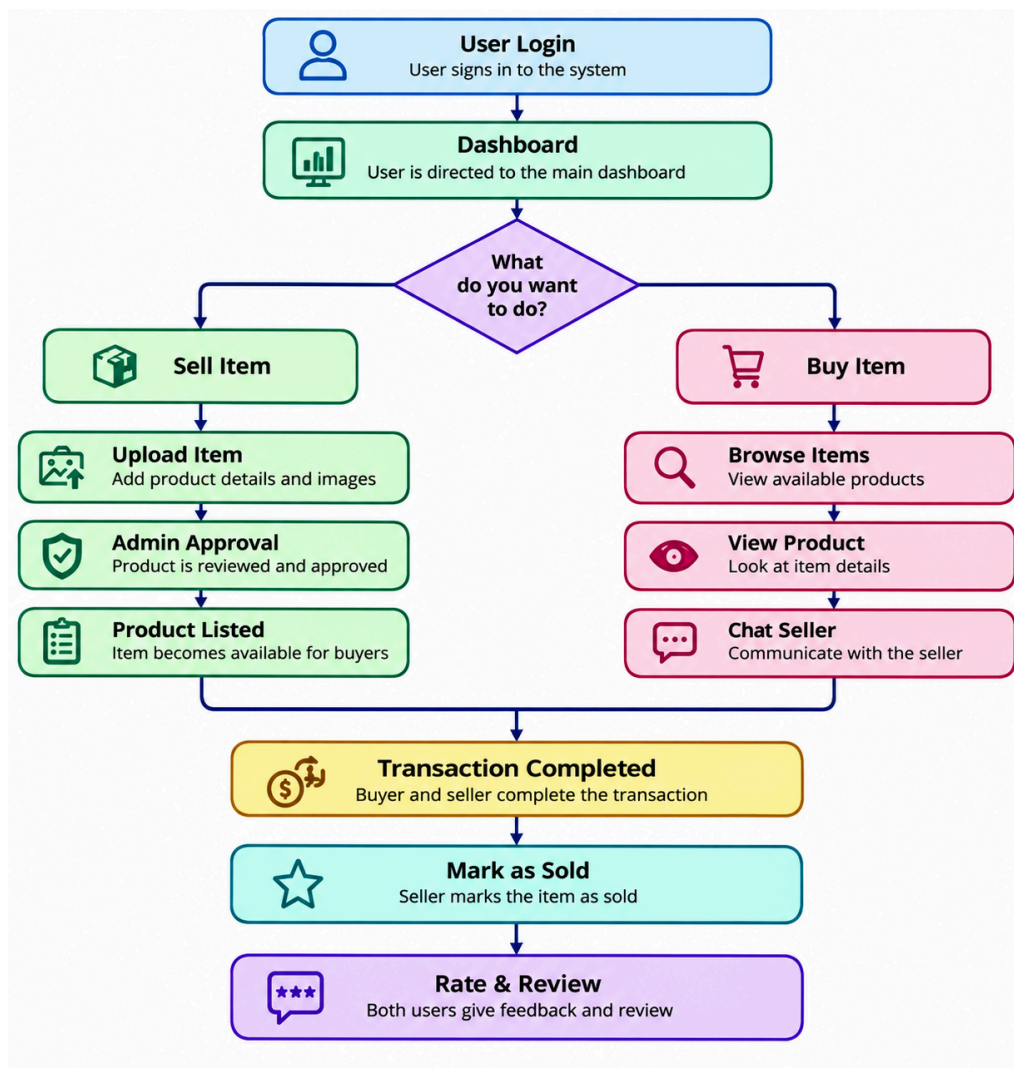
- Responsive Design
- Secure Authentication
- Fast Search
- High Availability
- Data Security
- Scalable Architecture
- Easy Maintenance

9. Technology Stack

Layer	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	Supabase
Database	Supabase
Authentication	Supabase Authentication

Image Storage	Supabase Storage
Notifications	Supabase Cloud Messaging (FCM)
Hosting	Supabase Hosting
Charts	Chart.js
Version Control	Git & GitHub & Vercel

10. System Workflow



11. Expected Benefits

For Students

- Buy affordable second-hand items
- Sell unused belongings quickly
- Safe transactions within the campus community

For Staff

- Convenient way to exchange useful items
- Trusted buyers and sellers
- Less dependency on informal communication channels

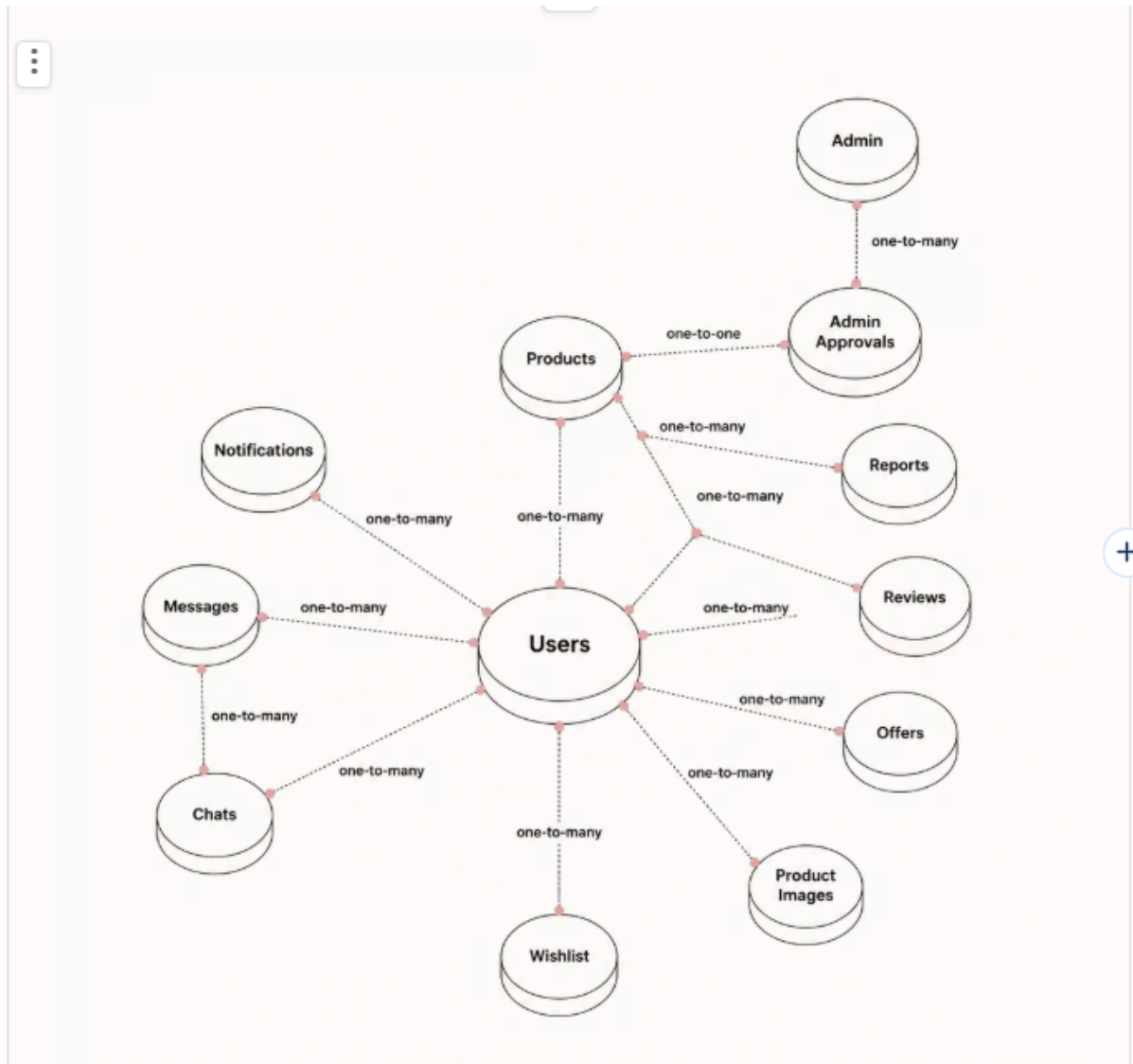
For College Administration

- Centralized and monitored marketplace
 - Reduced misuse compared to public platforms
 - Insights through analytics and reports
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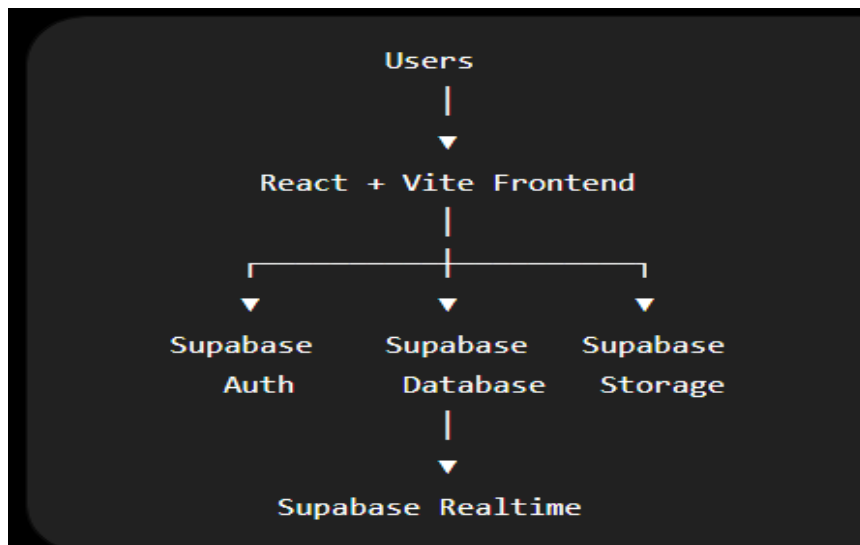
12. Why This Solution?

- ✓ Solves the lack of a dedicated campus marketplace.
 - ✓ Restricts access to verified college users, increasing trust.
 - ✓ Enables every user to act as both a buyer and a seller.
 - ✓ Reduces waste by encouraging reuse of products.
 - ✓ Improves efficiency with search, chat, notifications, and analytics.
 - ✓ Uses Supabase services for a scalable, secure, and low-maintenance solution suitable for a college project or hackathon.
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ER DIAGRAM:



FRONTEND - BACKEND ARCHITECTURE



Conclusion

The **Campus Marketplace Portal** provides a secure, centralized, and user-friendly platform for college students and staff to buy and sell used items within the campus community. It replaces unorganized methods such as WhatsApp groups, notice boards, and word-of-mouth communication with a reliable digital marketplace.

By using **Supabase Authentication, Database, Storage, and Realtime**, the system supports secure user access, product management, image storage, real-time buyer–seller communication, and efficient data management. Features such as product search and filtering, wishlist, notifications, product status tracking, admin moderation, reporting, and analytics improve the overall user experience and platform reliability.

The project promotes the reuse of products, reduces waste, helps users buy affordable items, and allows sellers to easily find interested buyers. Overall, the Campus Marketplace Portal offers a practical, scalable, and sustainable solution that makes campus buying and selling more convenient, transparent, secure, and efficient.